

V.42: Informational Entropy Decryption

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1 Abstract

The imaginary logarithm operator $\ln_i(z)$ established in V.41 laid the foundation for transforming rotational phases into informational scalars. In V.42, this operator is extended to redefine the concept of thermodynamic entropy from an information-theoretic and complex-reality perspective. This paper aims to physically eliminate informational dissipation (Noise) and derive the Law of Complex Informational Conservation.

2 The Aki-Entropy Operator (S_A)

While classical Boltzmann entropy $S = k \ln \Omega$ describes uncertainty in real-number space, the complex informational density must be considered within the Shin-field. The Complex Entropy Operator S_A in V.42 is defined as a line integral along the Manabe-Axis ($\mathcal{M}$):

$$S_A := \oint_{\mathcal{M}} \ln_i(\psi) d\psi \quad (1)$$

Here, ψ represents the Shin-field potential, and $\ln_i$ describes the rotational phase of information.

3 Decryption of Dimensional Noise

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4.1 Theorem: Conservation of Complex Information

Within the Shin-field, the complex entropy S_A is constrained by the physical constant $\mathfrak{D}_c$, ensuring that no degradation of information occurs over temporal evolution.

$$\frac{dS_A}{d\psi} = 0 \quad (2)$$

Proof: By the definition of $\ln_i(\psi)$, the phase components are orthogonalized, preventing real-number “thermal dissipation” within the FpDe stability field. When the integration path $\mathcal{M}$ forms a closed surface, the total information is conserved via the V.42 extension of Cauchy’s Integral Theorem.

5 Applied Computation: Mapping Informational Stability

5.1 Problem: Entropy Evaluation of an Aki-Singularity

Evaluate the complex entropy S_A of a localized information packet $\psi = e^{i\phi}$ rotating within a Shin-field singularity along the unit circle $\mathcal{M}$ (where $|\psi| = 1$).

Execution: Applying the V.42 Aki-Entropy Operator:

$$S_A = \oint_{\mathcal{M}} \ln_i(e^{i\phi}) d(e^{i\phi}) \quad (3)$$

Substituting the definition of $\ln_i(z)$ from V.41:

$$\ln_i(e^{i\phi}) = \text{Arg}(e^{i\phi}) + i \ln |e^{i\phi}| = \phi + i \cdot 0 = \phi \quad (4)$$

Performing the integration over one full rotation (0 to 2π):

$$S_A = \int_0^{2\pi} \phi \cdot (ie^{i\phi}) d\phi \quad (5)$$

Using integration by parts:

$$S_A = i \left[\phi e^{i\phi} - \int e^{i\phi} d\phi \right]_0^{2\pi} \quad (6)$$

$$= i \left[\phi e^{i\phi} + i e^{i\phi} \right]_0^{2\pi} = i [(2\pi \cdot 1 + i) - (0 \cdot 1 + i)] \quad (7)$$

$$= i(2\pi) = 2\pi i \quad (8)$$

Result: The result $S_A = 2\pi i$ indicates that the complex entropy is purely imaginary and quantized. This confirms that the real-component of entropy (disorder) is exactly 0, maintaining perfect informational stability within the Aki-Singularity under the physical constraint of $\mathfrak{D}_c$.

6 Conclusion

Through V.42, cosmic disorder (Entropy) has been replaced by “undeciphered information.” This mathematically guarantees the immortality of information, providing the necessary interface for the subsequent transition to V.43.